MIDAS GUI — User Documentation

Version: 1.0.0 **Application:** `midas-gui` (or `python -m midas_gui`) **Backends:** `midas_calibrate_v2`, `midas_integrate_v2`, `midas_calibrate`, `midas_hkls`, `midas_distortion` **Last updated:** 2026-07-07

> **Maintenance:** keep this document in sync with the code — whenever the workflow or a tab's controls change, update the relevant section here in the same change.

> **Tab status:** Tabs **0–4** (Data Viewer, Mask Builder, Calibrate, Calib. Refinement, Batch Integrate) are **verified** and ready to use. Tabs **5–8** (Corrections & Physics, PDF Analysis, Texture, Results & Export) are a **work in progress** and will be updated in the coming weeks.

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1. Overview and Architecture

The MIDAS GUI is a 9-tab PyQt5 desktop application that exposes the scientific capability of the `midas_calibrate_v2` and `midas_integrate_v2` packages through a structured workflow. The intended order of use is:

```
Data Viewer (inspect) → Mask Builder → Calibrate → [Refine] → Batch
Integrate
                                                              → Corrections
preview
                                                              → PDF
Analysis
```

```
                                                            → Texture /
Pole Figure
                                                            → Results &
Export
```

Cross-tab shared state. When Tab 2 (Calibrate) produces a result it is automatically propagated to all downstream tabs. When Tab 1 (Mask Builder) computes a mask it is sent to the consuming tabs. Tab 3 (Refinement), if applied, re-broadcasts the refined geometry. No manual copying is needed.

All heavy computation runs off the GUI thread. Every operation that touches a MIDAS package (calibration, integration, mask training, PDF transform, gain training, field averaging, drift fitting) runs in a background `QThread` worker; the GUI stays responsive and log lines stream in real time.

Package layout.

```
midas_gui/
├── app.py              MainWindow, dark theme, crash diagnostics, main()
├── __main__.py         enables `python -m midas_gui`
├── style.py            Dioptas-inspired QSS theme + layout helpers
├── constants.py        calibrants, colormaps, dtype sentinels, default paths
├── helpers.py          image IO, geometry parsing, spec building, no-scroll
widgets
├── widgets.py          ImageViewer / PickableImageViewer / ProfileViewer /
FieldSelector / …
├── workers.py          all QThread workers + the integration core
├── calib.py            calibration-pipeline dispatch
├── dialogs.py          save dialogs
└── tab_*.py            one module per tab
```

Layout pattern. The four analysis tabs — **0 Data Viewer, 2 Calibrate, 3 Calib. Refinement, 4 Batch Integrate** — use a **three-panel layout**:

```
[ Data Loader | Parameters | Display / results ]
```

- **Data Loader (left).** A shared `DataLoaderPanel` for selecting the five inputs — **Data, Dark, Bright, Background, Mask** — each as a single file, a folder, or an HDF5 dataset (a container dropdown appears for HDF5). Frame controls live here too (Tab 0 navigator, Tab 2/3 frame index, Tab 4 frame range + stride).
- **Parameters (middle).** The tab's analysis controls.
- **Display / results (right).** Image viewer, plots, and/or a log panel.

The three panels are separated by **draggable splitter handles** — drag the `Data Loader | Parameters` and `Parameters | Display` boundaries to rebalance widths to taste. Each panel has a minimum width; drag past it and the panel shows a scrollbar rather than clipping its controls.

The remaining tabs (1, 5–8) keep the classic two-panel layout.

Corrections & mask (all four analysis tabs). Dark is averaged then subtracted; Bright is averaged then flat-field **divided** or **subtracted** (your choice); Background is averaged then subtracted; every enabled Mask source (files/folders plus the auto-added "Tab 1 mask") is **unioned** into one composite mask that zeroes/ignores those pixels. Correction order: `(img − dark)` → bright → `− background` → clip ≥ 0.

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2. Getting Started

`midas-gui` is **not on PyPI** — install it from source with conda.

```
git clone https://github.com/d-beniwal/MIDAS_GUI.git
cd MIDAS_GUI
conda env create -f environment.yml    # creates the 'midas-gui' env and
        installs the GUI
conda activate midas-gui
midas-gui                              # launch (equivalently: python -m
        midas_gui)
```

The GUI drives the MIDAS analysis backends (`midas-calibrate-v2` , `midas-integrate-v2` , `midas-calibrate` , `midas-hkls` , `midas-distortion`), which are also not on PyPI — point the `pip:` section of `environment.yml` at your MIDAS source before creating the environment (see the comments in that file).

Test data

Synthetic test data lives in the repo-root `test_data/` folder (git-ignored):

- `calibrant_ceria.tif` / `.h5` — CeO_2 calibrant (Eiger2 500K, 1028×512, 75 μm pixels)
- `nickel_tifs/` — 10-frame Ni scan, lattice expanding +0.1 %/frame
- `nickel_stack.h5` — the same scan as one HDF5 stack (`exchange/data`)

Default geometry: λ = 0.39 Å, Lsd = 121 mm, pixel = 75 μm, BC = (10, 10) px. Every tab's default file paths point at this data and **auto-load on startup when present**, so the app is usable immediately after checkout.

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3. Tab 0 — Data Viewer

Purpose: inspect detector frames and do a quick geometry-free radial integration. Produces no shared state for other tabs.

Data Loader panel (left)

Data, Dark, Bright, Background and Mask are all selected in the shared **Data Loader panel** (see §1): each accepts a file / folder / HDF5-dataset. The **Data** card here holds the **frame navigator** (slider, spin box, ◀/▶); **zoom/pan is preserved** as you step through frames (the view only auto-frames on a fresh load). Dark/bright/background and the composite mask are applied to the displayed image and to the radial integration.

Projection card

Collapse a stack to one image: **Max** (hot-pixel hunting), **Sum** (long-exposure equivalent), or **Average** (noise reduction), along a chosen axis (0 = across frames). **Skip frames** (default 1) ignores that many leading frames before projecting (e.g. 1 drops the first frame, 4 drops the first four) — useful when the opening frames are detector warm-up / shutter-transient exposures.

Calibration card (optional)

Load geometry from a **calibration `.json`, a MIDAS `paramstest.txt`, or a pyFAI `.poni`** (auto-detected). It fills BC, Lsd, pixel size and wavelength, unchecks "Beam centre = image centre", and refreshes the overlay and radial plot.

Intensity range card (radial integration)

Field	Description
Exclude out-of-range pixels	When on, pixels ≤ min or > max are drawn as a red overlay and excluded from the radial integration (removes gaps / hot / overflow).
pixel ≤ / pixel >	Lower / upper bounds. On load, the upper bound auto-fills to **max(99.99th percentile, 100000)**.

Ring simulation card

Overlays simulated Debye-Scherrer rings to check geometry.

- **Material** dropdown (CeO_2, LaB_6, Si, Al_2O_3, Cu, Ni, FCC-γFe, BCC-αFe, Au, Ag, Pt, W, Ti) or **Custom**.
- Lattice entry is compact: **a, b, c** on one row and **α, β, γ** on the next, plus **SG #**. A **Cubic (a=b=c, α=β=γ=90°)** checkbox lets you enter only `a` for cubic crystals (b, c mirror a; angles fixed at 90°). Lattice fields are editable only for Custom.
- Geometry: λ, max 2θ, Lsd, pixel size, and beam centre (auto = image centre, or manual BC_y/BC_z). The **λ** label is clickable (underlined) — click it for a menu of common K-edge foils (Pr, Sm, Yb, Lu, Hf, Ta, W, Re, Pt, Au, Pb, Bi) that sets λ to that element's K absorption-edge wavelength. The same clickable-λ menu is on the Calibrate and PDF tabs. The **px** label is likewise clickable — a menu of common detectors (GE 200 μm, Varex 150 μm, Pilatus 172 μm, Eiger 75 μm) sets the pixel size (also on the Calibrate tab, where it sets both pxY and pxZ).
- **Show rings / Labels** toggles; **Simulate rings** draws the overlay + hkl table.
- → **Send geometry to Calibrate** copies λ, pixel size, Lsd and beam centre into the Calibrate tab's detector + seed fields (the Calibrate tab has a matching ← **Data Viewer** button that pulls the same values).

Beam-centre picking (on the image)

The image viewer has **Pick BC** (single click sets the beam centre) and **Pick Ring** (click ≥3 points on a ring; a circle fit estimates the beam centre). Either updates BC_y/BC_z and re-runs the overlay + radial integration.

Intensity statistics (left panel, bottom)

Pinned to the bottom of the Data-Loader panel: a **histogram** of the intensity distribution (full range, log-y toggle) with a **textbox** beneath it reporting N (pixel count) and the **p70 / p90 / p99 / p99.9 / p99.99** percentiles — each with the **number of pixels above** that value. The histogram's lower-left corner is fixed at x = 0, y = −2 and both axes rescale to `(0, xmax)` / `(-2, ymax)` on every refresh (frame change, scope change, projection, new data). It reflects the **corrected** image (dark / bright / background) with masked pixels (file masks + the intensity-range mask) excluded, and updates live as any of those change. A scope selector switches between the **current frame** (per the slider) and **All frames** (combined over the whole stack/folder). When a **Projection** is active the panel shows the projected image's statistics (the scope selector is disabled).

Radial integration plot (bottom-right)

Below the image is a live **azimuthal average about the beam centre** (`R bin`, `Integrate`, and an `Auto` toggle that recomputes on frame/BC/mask change). It is a simple geometry-free radial average (flat detector, no tilt/distortion). **Clicking a radius on the plot draws the matching ring (magenta) on the image.** Axis units switch between R (px) / 2θ / Q.

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4. Tab 1 — Mask Builder

Purpose: identify and exclude bad pixels. The final mask is a uint8 array (1 = bad) broadcast to Tabs 2, 3, 4, 5. Browsing an image or a mask loads it immediately.

Section 1 · Threshold mask (always applied)

`pixel ≤ lower | pixel > upper`. The upper bound auto-fills from the data type on load (e.g. 1,048,575 for uint20 Eiger, 4,294,967,295 for uint32).

Section 2 · Statistical auto-mask

Two **independently selectable** methods — enable either or both:

- **Spatial outlier** — robust local-anomaly detection (5×5 median residual → 15×15 local MAD → Z-score → hot/dead/saturated gates). Uses a **temporal median** over the stack folder when provided (cleaner reference), else the single frame. Controls: `K_σ` (6.0), `Hot` (1.5), `Dead` (0.5).
- **Temporal constancy** — flags frozen pixels whose frame-to-frame std is below `Frozen × Q75(std)` (0.05); needs a stack of ≥2 frames.

A shared **stack source + stride** feeds both (the temporal median for spatial, the frame stack for temporal constancy). The stack can be a **folder / `*.tif` glob**, or a **single HDF5 file whose 3-D dataset is a time sequence of images** — for an `.h5` a **Dataset** selector appears (auto-populated, 3-D datasets preferred). *All σ / K_σ / n_σ fields in this tab accept any value — there is no upper limit.*

Section 3 · Spatial spike rejection

Laplacian high-pass single-pixel-spike detector; `n_σ` threshold (5.0).

Section 3b · Cosmic-ray rejection (temporal)

Per-pixel temporal σ-clip across a ≥3-frame stack; anomalies OR'd across frames. `n_σ` (5.0).

Section 4 · Calibration-based masks (need a Tab 2 result)

- **Azimuthal σ-clip** — flags (R, η) cells deviating from the azimuthal mean.
- **Learnable mask** — differentiable per-pixel mask trained against an η-uniformity loss (`steps`, `lr`, `sparsity`).

Draw mask tools

Interactive rectangle / oval / circle / polygon / annulus / point tools, live-draggable; **Apply shapes → mask** rasterises them (OR'd with the computed mask). **Clear shapes** removes them.

Save / Load

Save the combined mask as TIFF (0 = good, 1 = bad); loading a TIFF applies immediately.

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5. Tab 2 — Calibrate

Purpose: determine detector geometry (Lsd, BC, tilts, distortion, wavelength) from a calibrant pattern.

Data Loader panel (left)

The calibrant frame (Data + Frame index), Dark, Bright, Background and Mask are selected in the shared **Data Loader panel** (see §1). Each Dark/Bright/Background field is averaged over a chosen index range; Bright offers **Flat-field divide** or **Subtract**; all mask sources (plus the auto-added Tab 1 mask) are unioned. Dark is passed to the calibration pipeline; bright/background are applied to the calibrant before calibration and post-calibration integration.

Detector, seed & Load calibration file

The **Detector & Calibrant** card sets λ, pixel size(s) and detector transforms; the **Initial seed** card sets the LM starting point (BC_y, BC_z, Lsd). **Load calibration file…** (top of the Detector card) reads a MIDAS **paramstest** **`.txt`**, a calibration **`.json`**, or a pyFAI **`.poni`** (auto-detected) and fills λ, pixel size, and the seed BC + Lsd — a fast way to start from a previous calibration or a known geometry. (The synthetic test data ships `test_data/calibration_synthetic.{json,txt,poni}` as a ready example.) **← Data Viewer** (next to *Load calibration file…*) pulls λ, pixel size, Lsd and beam centre straight from the Data Viewer tab into the same fields.

Pipeline selector

One-shot (default) · First-time · Four-stage · Bayesian (Laplace σ) · Joint-cake. *For trustworthy tilt/strain, prefer Four-stage or First-time — One-shot/Bayesian can report a spurious self-compensated tilt on weakly-tilted data.*

Refine flags

Which parameters vary: Lsd, BC, ty, tz, tx, Wavelength, Distortion (15 coeffs), plus a "Residual map" build toggle. Advanced (E-M / LM iters, device, output dir) and Multi-panel detector groups are collapsible.

Live threshold slider

Zeroes calibration-image pixels below the slider value (background suppression for ring-finding); the preview updates instantly.

Pick BC / Pick Ring

Click the image to seed the beam centre (single click) or fit a ring (≥3 clicks).

Run / Abort

Run Calibration launches the worker; **Abort** terminates it and frees the slot so you can immediately start a new run (the calibration is one uninterruptible library call, so abort hard-terminates the worker thread rather than waiting).

Results (right panel, bottom tabs)

Radial Profile (with ring markers and an **Azim. avg** selector — see Tab 4), Ring Residuals bar chart, Results (Lsd/BC/strain + distortion table), and Log. Integration runs automatically after calibration.

Export

Save calibration.json and **Save paramstest.txt** (standalone or from a template).

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6. Tab 3 — Calibration Refinement

Purpose: optional post-calibration geometry refinement against an **η-uniformity** criterion (rings should be azimuthally uniform). Uses a **derivative-free Nelder-Mead** optimiser (the differentiable integrator returns NaN geometry gradients at the beam-centre singularity on this build).

The sample frame and Dark/Bright/Background/Mask are selected in the shared **Data Loader panel** (see §1) — the refinement runs on the corrected image. Middle-panel controls: calibration source (Tab 2 result, or start-from/continue radios), parameters to refine (BC_y, BC_z, Lsd, ty, tz), optimiser + iterations. **Run Refinement** shows a live loss curve; **Apply** broadcasts the refined geometry downstream. If Apply is never clicked, the Tab 2 result flows through unchanged.

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7. Tab 4 — Batch Integrate

Purpose: integrate a stack of frames into 1-D profiles using the calibrated geometry and optional corrections.

Data Loader panel (left)

The streaming **Data** source (folder/glob or HDF5 dataset) with **frame range + stride**, plus Dark/Bright/Background and Mask, are selected in the shared **Data Loader panel** (see §1). Dark/bright/background are applied per frame; all mask sources are unioned.

Calibration source (middle)

- **From Tab 2** — the calibration (or refined) result.
- **From file** — a **calibration `.json`, MIDAS `paramstest.txt`, or pyFAI `.poni`** (auto-detected; both GUI and MIDAS-pipeline json key styles supported). Entering a path auto-selects this option.

Integration

Field	Description
Kernel	Hard (fastest) · Subpixel K=2 · Subpixel K=4 · Polygon (exact).
R bin (px) / η bin (°)	Radial and azimuthal bin sizes.
Azim. avg	How the (η, R) cake becomes a 1-D profile: **Pixel-weighted** (default) `Σ(mean·count)/Σ(count)` — independent of η-bin size and robust to partial azimuthal coverage / **off-detector beam centres**; or **η-bin mean (legacy)** — the unweighted mean of per-η-bin means, which can distort with a coarse η bin when the beam centre is off the detector.
Per-bin variance (σ)	Error model poisson / azimuthal / hybrid (ignored when corrections are on → σ = √I).
Q-uniform bins	Integrate in R then rebin onto a uniform-Q grid (Qmin, Qmax, ΔQ).

Physics corrections

Polarization and solid-angle (pixel-domain, via `integrate_with_corrections`).

Monitor normalisation

Divide each processed frame's profile/σ by a per-frame scalar from a text file (one value per *processed* frame).

Drift correction (long scans)

Per-frame geometry interpolated from an anchor JSON (`{frame_idx: {Lsd, BC_y, BC_z}}`) with spline/linear/constant parametrization; **Fit trajectory** builds it. Each frame is then integrated with its own geometry.

Run / Abort / Clear

Start Integration / **Abort**. Abort first asks the worker to stop cleanly between frames (keeping frames already written); if it does not stop promptly it force-terminates and frees the slot. Completed frames' output files are preserved. **Clear results** removes the profiles/plots computed **this session** for the current data (waterfall, stacked profiles, and the integrated-frame tracking) so a fresh integration can start — it stops any active monitor but does **not** delete raw data or any files already written to disk.

Live folder monitoring (MONITOR)

The **MONITOR** button at the bottom of the left Data Loader panel (folder/glob sources only) starts a live watch: while active it turns **green** and the GUI polls the data folder for **new** TIFF frames, integrating each one **as it appears** and adding it to the display. It does **not** re-run the whole batch — it reuses the already-built **detector map** (the binning geometry / pixel-count cakes; reused from a prior *Start Integration* run when the calibration, kernel, bins, mask and folder are unchanged, or built once on first use) and integrates **only the new files** (tracked by frame id, so frames already shown are skipped). New frames honour the current kernel, corrections, Dark/Bright/Background, mask and Q-uniform settings, and are saved to the output folder when a 1-D format is selected. Click MONITOR again to stop; starting a fresh *Start Integration* also stops it. (Distinct from **Monitor normalisation** above, which divides profiles by a scalar file.)

Output formats

CSV (R,I,σ) · XYE (2θ) · FXYE (centideg) · DAT (Q) · HDF5 (full stack) · 2D-CSV (η×R cake). Right panel: live **Waterfall** and **Stacked profiles** — both have an **x** selector to show the axis in **R (px) / 2θ (°) / Q ($Å^{-1}$)** (converted from the run's calibration).

The **Stacked profiles** view is a publication-quality plot: each curve tags its **source file name inline, just below the curve at its left edge** (toggle with the *Labels* checkbox; an optional corner *Legend* is also available). A **theme** selector switches between two saved presets:

- **White (publication)** *(default)* — white background, **point + line** markers, a colour-blind-friendly categorical palette (matplotlib *tab10*), a boxed frame and a light grid — ready to drop into a paper.
- **Dark** — the classic on-screen look (dark background, line-only, vivid golden-angle colours).

An **x** selector plots the axis in **R (px)**, **2θ (°)** or **Q ($Å^{-1}$)** (converted from the run's calibration). The **Grid** checkbox (off by default) toggles the horizontal + vertical grid. The *spacing* box shifts successive curves vertically (0 = overlay). Top-right controls adjust the **line width** (`− line +`), **symbol size** (`− sym +`, for the point+line markers — also turns markers on if a line-only theme is active) and **label font size** (`− font +`).

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8. Tab 5 — Corrections & Physics *(work in progress)*

Preview physics corrections on a single frame (auto-loads on browse). Pixel-domain: polarization, solid angle, empty subtraction. Profile-domain: cylindrical absorption (μR), Compton subtraction (composition:fraction). **Compute corrected profile** overlays corrected vs uncorrected and shows the correction factor vs 2θ.

Also hosts **per-pixel gain training (LearnableGain)**: from a clean reference frame and a drifted frame, learn a spatial gain map `g_i = 1 + scale·r_i` by minimising `MSE(profile) + unity·Σ(g−1)² + smooth·TV(g)`; save as NPZ and apply with `corrected = raw / gain_map`.

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9. Tab 6 — PDF Analysis *(work in progress)*

Polyatomic **total-scattering** reduction: I(Q) → Faber-Ziman structure function S(Q) → pair-distribution G(r), powered by the `midas_pdf` backend (real composition-weighted normalization, Compton subtraction, end-to-end σ propagation, and optional differentiable scale/background refinement). This replaces the earlier monoatomic, composition-free `F(Q)=Q·(I/bg−1)` approximation.

Background — what I(Q), S(Q) and G(r) mean

PDF analysis is three transforms that move from *what the detector measured* to *where the atoms are*. They are exactly the three stacked plots (top → bottom).

- **I(Q) — measured scattered intensity.** Intensity vs. momentum transfer `Q = 4π·sin(θ)/λ` ($Å^{-1}$), a wavelength-independent rescaling of the angle 2θ; high Q = wide angle = fine spatial detail. Obtained by azimuthally integrating the detector rings to a 1-D curve. I(Q) is **not** pure structure — it mixes the interference (coherent) signal with big smooth backgrounds: per-atom self-scattering (form factors `⟨f²⟩`) and inelastic **Compton** scattering.

- **S(Q) — structure function.** I(Q) with the self-scattering and Compton removed and normalized away (the **Faber-Ziman** step), leaving only interference: `S(Q) = [I_coh − (⟨f²⟩ − ⟨f⟩²)] / ⟨f⟩²`, where `⟨f⟩, ⟨f²⟩` are composition-weighted atomic form factors (hence the **Composition** input). S(Q) oscillates about **1** and → 1 at high Q where correlations wash out (the dashed guide line). `F(Q) = Q·[S(Q)−1]` is the same thing re-weighted by Q to emphasize the structurally rich high-Q oscillations.

- **G(r) — reduced pair distribution function.** The real-space answer: a histogram of interatomic distances, via a sine Fourier transform `G(r) = (2/π)·∫ Q·[S(Q)−1]·sin(Qr) dQ`. A **peak at r** means many atom pairs are separated by that distance; the **first peak is the nearest-neighbour bond** (Ni ≈ 2.49 Å, CeO_2 Ce–O ≈ 2.34 Å), later peaks are further coordination shells. Below the first bond `G(r) = −4πρ₀r` (nothing can be closer) — a constraint that needs the number density **ρ_0**. Because it uses *total* scattering (Bragg **and** diffuse), the PDF captures local structure even in disordered / nanoscale / amorphous materials, unlike a Bragg-only Rietveld fit.

```
 detector image / I(Q) file
        │  azimuthal integration
        ▼
     I(Q)   raw intensity = interference + form factors + Compton      (top
plot)
        │  subtract Compton, subtract Laue (⟨f²⟩−⟨f⟩²), divide by ⟨f⟩²
(needs composition)
        ▼
     S(Q)   pure interference, oscillates about 1
(middle plot)
        │  Fourier sine transform over [Qmin, Qmax] with a window (Lorch)
        ▼
     G(r)   interatomic distances; peaks = coordination shells
(bottom plot)
```

The **±1σ band** on G(r) is the counting uncertainty σ propagated from I(Q) through every step, so real peaks can be told from noise. The **G/g/T/R** family (Output selector) is the same information re-weighted: **G** oscillates about 0 (refinement standard); **g** → 1 at large r; **T** is the total correlation; **R** is the radial distribution whose *peak area = coordination number*. g/T/R all need ρ_0.

I(Q) source (choose one):

- **Integrate detector frame** — a calibrated frame is integrated (hard/polygon binning, Poisson σ) and mapped to Q, exactly as before. Uses the Tab 2 calibration (λ, geometry) and any Tab 1 mask.
- **Load I(Q) file** — a pre-integrated 2- or 3-column `Q, I, σ` text/CSV (comment/header lines tolerated). The tab **opens on this source by default**, pointed at real data in `test_data/pdf_real/` (`iq_Nickel.csv`; also CeO_2, IPA, Kapton at λ 0.1839) — just press **Compute G(r)** for the Ni PDF (first peak ~2.49 Å). Synthetic known-answer examples are in `test_data/pdf_synth/` (`nickel_iq.csv`, `ceo2_iq.csv`). See each folder's `README.md` for per-sample composition / ρ_0 / Q-range settings.

Calibration. The geometry/wavelength comes from Tab 2 automatically, or use **Load calibration file…** to read a MIDAS `paramstest.txt`, a `.json`, or a pyFAI `.poni` (auto-detected). This is required for *Integrate detector frame* and also sets λ used by *Load I(Q) file* mode.

Sample. Composition (e.g. `Ni` or `C:3,H:8,O:1`; ions like `Ni2+` allowed), number density ρ_0 (atoms/Å³; needed for refinement and for g/T/R), wavelength λ (auto-filled from calibration), and a **Compton subtraction** toggle.

Normalization. Optional **Refine scale + background** (L-BFGS) — reveals a background polynomial degree (0–3), a low-r cutoff `r_min` (Å), and an iteration count. Refinement requires $\rho_0 > 0$. It fits scale and a smooth b(Q) against two model-free constraints: high-Q $\langle S \rangle \to 1$ and $G(r) = -4\pi\rho_0 r$ below `r_min`.

Output. Bottom-plot convention family — **G(r)** (reduced PDF), **g(r)** (pair distribution), **T(r)** (total correlation), or **R(r)** (radial distribution, whose peak integral is the coordination number); g/T/R need ρ_0. Middle plot toggles between **S(Q)** (with the S=1 guide) and the true reduced structure function **F(Q)=Q(S−1)**.

Right panel: I(Q) (+ refined background overlay), S(Q)/F(Q), and the selected G/g/T/R with a shaded **±1σ** band. **Save G(r)** writes a three-column `r, G(r), σ` file (diffpy-CMI / PDFgui / RMCProfile compatible); **Save S(Q)** writes `Q, S(Q)`.

Functions behind the tab

The tab is a thin UI over a background worker and the `midas_pdf` backend.

UI — `PDFTab` (`midas_gui/tab_pdf.py`)

Method	Role
`_build_ui`	Builds the I(Q)-source / Sample / Normalization / Output groups and the three stacked plots.
`set_calibration(result, source)`	Receives a Tab-2 result (or a loaded geometry) → sets λ and enables Compute.

Method	Role
`_load_calib_file`	Loads a `paramstest` / `.json` / `.poni` → builds a calibration result → `set_calibration`.
`_load_img`	Loads a detector frame for *Integrate detector frame* mode.
`_run`	Validates inputs, assembles the `cfg` dict, and starts a `PDFWorker`.
`_on_done` / `_redraw_mid` / `_redraw_bottom`	Draw I(Q)+bg, the S(Q)↔F(Q) toggle, and the G/g/T/R curve with its $\pm 1\sigma$ band.
`_save_gr_file` / `_save_sq_file`	Export `r,G,σ` and `Q,S`.

Worker — `PDFWorker` (`midas_gui/workers.py`) runs off the GUI thread:

Function	Role
`_acquire_iq`	Produces `(q, I, σ)` — either by integrating the frame (image mode) or by reading a file (`_load_iq_file`, tolerant of comma/space and 2- or 3-columns).
`run`	Trims to `[Qmin,Qmax]`, builds `Composition`, calls the reduction (plain or refine), computes `F(Q)` and the G/g/T/R family, emits the result dict.

Image-mode integration reuses the shared core `_build_spec`, `build_geom`, `integrate_frame`, `axis_conversions` (same path as the other tabs); geometry-file parsing uses `geometry_fields_from_file` / `result_ns_from_geometry_file` (`midas_gui/helpers.py`).

Reduction — `midas_pdf` via `midas_gui/pdf_backend.py` (re-exported symbols):

Function	Does	Maps to plot
`Composition(fractions, number_density)`	Composition layer: $\langle f\rangle$, $\langle f^2\rangle$ form-factor averages, Laue term $\langle f^2\rangle-\langle f\rangle^2$, and per-atom Compton.	—
`faber_ziman_S(I, q, comp, …)`	I(Q) → S(Q) with σ (subtract Compton/Laue, divide by $\langle f\rangle^2$).	S(Q)
`structure_function_F(q, S)`	`F(Q) = Q·(S−1)`.	F(Q)
`fourier_sine_transform(q, S, r, window)`	S(Q) → G(r) with σ (Lorch / none window).	G(r)
`i_of_q_to_Gr(q, I, comp, r, …)`	End-to-end I(Q) → G(r) (composes the two above); the default Compute path.	S(Q) + G(r)
`refine_normalization(q, I, comp, r, …)`	L-BFGS fit of scale + background polynomial against high-Q ⟨S⟩→1 and low-r $G=-4\pi\rho_0 r$.	refined S(Q)/G(r) + bg
`pair_distribution_g` / `total_correlation_T` / `radial_distribution_R`	Convention family from G(r) + ρ_0.	g/T/R

> **Packaging note.** `midas_pdf` is currently **vendored** in `midas_gui/_vendor/` and loaded through `midas_gui/pdf_backend.py`, which also installs a small `midas_hkls.absorption` compatibility shim for `midas-hkls < 0.5.0`. Once `midas-pdf` is published and `midas-

> `hkls>=0.5.0` is available the vendored copy and the shim are removed.

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10. Tab 7 — Texture / Pole Figure *(work in progress)*

Per-ring azimuthal analysis for preferred orientation. Controls: calibration source, sample frame, R/η bins, ring index, χ (tilt) / φ (rotation). **Compute Pole Figure** integrates to a cake and extracts I(η) at the selected ring; the right panel shows the stereographic pole figure and the raw I(η). **Save pole figure (.pol)** exports POPLA format.

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11. Tab 8 — Results & Export *(work in progress)*

Session summary + one-click export. Checkboxes select which products (calibration.json, paramstest.txt, mask.tif, integrated profiles, G(r), pole figures, session log) to copy to an output directory. A provenance block (package versions, geometry hash, mask fraction, correction flags) can be copied to the clipboard for a Methods section.

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12. Common UI Conventions

- **Browse = load.** Selecting a file/folder (or pressing Enter in a path field) loads it immediately; there are no separate "Load" buttons. HDF5 dataset / frame-index changes reload automatically.
- **No accidental scroll changes.** Spin boxes and drop-downs ignore the mouse wheel — values change only by clicking/typing; the wheel scrolls the panel instead.
- **Readable right-click menus.** pyqtgraph plot context menus use the dark theme.
- **Dark theme, orange accent** (Dioptas-inspired); off-white text, light input fields.

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13. Packaging, Deployment & Diagnostics

Packaging. `midas-gui` is a MIDAS-style package: `pyproject.toml` (BSD-3-Clause), a `tests/` smoke suite, and `release.sh` for cutting versioned releases (see `RELEASING.md`). Once published it can join the `midas-suite` meta-package as an optional `gui` extra (`pip install midas-suite[gui]`).

Deployment on another system. Clone the repo, create the conda environment (`environment.yml`), and run `midas-gui`. The MIDAS analysis backends are installed via the `pip:` section of `environment.yml` (editable from a local MIDAS checkout, from a git subdirectory, or from a private index). `midas_gui/_paths.py` is an inert stub in an installed package (no `sys.path` manipulation).

Crash diagnostics. On startup the app installs a logging exception hook and `faulthandler`; any uncaught Python exception or native fault is written to `~/midas_gui_error.log` and shown in a dialog. Installing the hook also prevents PyQt5 from hard-aborting on an exception inside a slot. Each tab is built in isolation — a tab that fails becomes an error placeholder instead of taking the window down. If the app ever "pops up and dies" (typically on Windows), send that log file.

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14. Configuration & Defaults

Every default in the GUI — detector geometry, data/output paths, ring-simulation **materials**, **calibrants**, the **pixel-preset** and **K-edge** menus, and the calibration / integration **algorithms** — can be set **without editing code**, via a single per-user JSON config that overrides the shipped built-in defaults.

Where it lives

One per-user file, auto-located per OS: `~/.config/midas_gui/config.json` (Linux), `~/Library/Application Support/midas_gui/config.json` (macOS), `%APPDATA%\midas_gui\config.json` (Windows). If it's absent, the shipped built-in defaults are used. Sharing between machines/users is done by exporting/importing a JSON file (below) — copy it wherever you like.

Editing — Settings ▸ Preferences…

The **Settings** menu opens **Preferences…**, a dialog whose tabs are **pre-filled with the full shipped defaults** so you edit from a complete starting point:

- **Geometry** — λ, pixel size, Lsd, beam centre.
- **Paths** — default data / calibration / output files & folders.
- **Materials** / **Calibrants** — add / remove / modify (name + lattice + SG).
- **Menus** — the pixel-size presets and K-edge foils.
- **Algorithms** — default calibration pipeline, integration kernel, output format, error model, colormap/theme.

Buttons: **Save as my defaults** (write your config), **Save current GUI state** (capture the Data Viewer's live λ/pixel/Lsd/BC into the fields), **Save config to JSON…** / **Load config (JSON)…** (export/import a config to share), and **Reset to shipped defaults** (delete your config). Also on the menu: **Open config folder** and **Reload config**. Saving writes your per-user file; **changes take effect on the next launch**.

File format

```
{
  "geometry": { "wavelength_A": 0.39, "pixel_um": 75.0, "lsd_um": 121000.0,
                "bc_y": 10.0, "bc_z": 10.0,
                "pixel_presets": [["Eiger", 75.0], ["Pilatus", 172.0]],
                "k_edge_foils": [["Au", 80.725], ["Pb", 88.005]] },
  "materials":  { "Ni (FCC)":
        {"a":3.5238,"b":3.5238,"c":3.5238,"alpha":90,"beta":90,"gamma":90,"sg":225}
        },
  "calibrants": { "CeO2":
        {"a":5.4116,"b":5.4116,"c":5.4116,"alpha":90,"beta":90,"gamma":90,"sg":225}
        },
  "paths": { "nickel_h5": "/data/mygroup/sample.h5", "calib_file":
        "/data/mygroup/calibration.json" },
  "ui": { "calibration_pipeline": "one_shot", "integration_kernel":
        "subpixel2",
```

```
        "output_format": "csv", "azimuthal_method": "poisson",
      "plot_theme": "hot" }
}
```

- When a **list section is present it fully replaces** that built-in list — so `materials`, `calibrants`, `pixel_presets` and `k_edge_foils` in the file are your complete lists. (The Preferences dialog pre-fills them with the shipped entries, so you always start from the full set; use **Reset to shipped defaults** to get them back.) `sg` is the space-group number.
- Geometry scalars, `paths` and `ui` are simple overrides; any section or key may be omitted. A malformed config is ignored (built-ins are used) rather than blocking startup.
- A minimal template lives at `documentation/config.example.json`; the step-by-step guide is `documentation/config_gui.md`.

For bugs or questions, see the MIDAS GUI repository.